Задача № 1

Дано $f: \mathbb{R}^n \to \mathbb{R}$ $\quad f(x) = \|x\|$ $\quad \|x\| = \sqrt{\langle x, x\rangle}$

$$\frac{\partial}{\partial x_i} = \sqrt{x^T x} = \frac{1}{2}\left(x^T x\right)^{-\frac{1}{2}} \cdot \left(x^T x\right)'_{x_i} =$$

$$\frac{1}{\|x\|}$$

$$\frac{\partial}{\partial x_i}\left(x^T x\right) = \left((x_1 \dots x_n)\begin{pmatrix} x_1 \\ i \\ x_n \end{pmatrix}\right)'_{x_i} = \left(x_i^2\right)' = 2x_i$$

$$\Rightarrow \nabla\left(x^T x\right) = 2x$$

$$\nabla f = \frac{1}{2} \cdot \frac{1}{\|x\|} \cdot 2x = \frac{x}{\|x\|}$$

$$H_{ij} = \frac{\partial}{\partial x_j}\left(\frac{x_i}{\|x\|}\right)$$

$i = j$ $\quad \frac{\partial}{\partial x_i}\left(\frac{x_i}{\|x\|}\right) = \frac{1 \cdot \|x\| - x_i \cdot \frac{x_i}{\|x\|}}{\|x\| \cdot \|x\|} = \frac{\|x\| - \frac{x_i^2}{\|x\|}}{\|x\|^2} = \frac{\|x\|^2 - x_i^2}{\|x\|^3}$

$i \neq j$ $\quad \frac{\partial}{\partial x_j}\left(\frac{x_i}{\|x\|}\right) = \frac{(x_i)'_j \cdot \|x\| - x_i \cdot (\|x\|)'_j}{\|x\| \, \|x\|} = \frac{-x_i x_j}{\|x\|^3}$ ($(x_i)'_j = 0$)

$$H_{ij} = \begin{cases} \frac{\|x\|^2 - x_i^2}{\|x\|^3}, & i = j \\ -\frac{x_i x_j}{\|x\|^3}, & i \neq j \end{cases}$$

Задача № 2

$f: \mathbb{R}^n \to \mathbb{R}$ $\quad f(x) = G(\langle c, x\rangle)$ $\quad c \in \mathbb{R}^n$

$G: \mathbb{R} \to \mathbb{R}$: $G(x) = \frac{1}{1 + e^{-x}}$

Найти $\nabla f(x)$:

$$\frac{\partial}{\partial x_i} G(x) = \frac{1}{1 + e^{-x}} = \left(1 + e^{-x}\right)^{-1} = -1\left(1 + e^{-x}\right)^{-2} \cdot \left(e^{-x}\right) \cdot (-1)$$

$$= \frac{e^{-x}}{\left(1 + e^{-x}\right)^2}$$

$(\langle c, x\rangle)'$

$$\Rightarrow \nabla f(x) = \frac{e^{-\langle c, x\rangle}}{\left(1 + e^{-\langle c, x\rangle}\right)^2} \cdot c$$

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∇H

$$\frac{\partial}{\partial x_j}\left(\frac{e^{-\langle c,x\rangle}}{(1+e^{-\langle c,x\rangle})^2}\cdot c_i\right) = \frac{c_i}{(1+e^{-\langle c,x\rangle})^4}\left[\left(e^{-\langle c,x\rangle}\right)'_{x_j}\left(1+e^{-\langle c,x\rangle}\right)^2 - e^{-\langle c,x\rangle}\left(\left(1+e^{-\langle c,x\rangle}\right)^2\right)'_{x_j}\right]$$

$$= c_i c_j\left(-\frac{e^{-\langle c,x\rangle}}{(1+e^{-\langle c,x\rangle})^2} + \frac{2e^{-2\langle c,x\rangle}}{(1+e^{\langle c,x\rangle})^3}\right) =$$

$$= \frac{e^{-\langle c,x\rangle}}{(1+e^{-\langle c,x\rangle})^3}\left(e^{-\langle c,x\rangle} - 1\right)\cdot c_i c_j$$

$$H = \frac{e^{-\langle c,x\rangle}\left(e^{-\langle c,x\rangle} - 1\right)}{(1+e^{-\langle c,x\rangle})^3} cc^T$$

Задача N 3

$f: \mathbb{R}^n \to \mathbb{R}$ $f(x) = \ln(1+e^{\langle a,x\rangle})$ $a \in \mathbb{R}^n$

$$\nabla f(x) = \frac{e^{\langle a,x\rangle}}{1+e^{\langle a,x\rangle}}\cdot a$$

$\nabla^2 f(x)$

$$\frac{\partial}{\partial x_i \partial x_j} = a_i\left(\frac{\partial}{\partial x_j}\frac{e^{\langle a,x\rangle}}{1+e^{\langle a,x\rangle}}\right) = a_i a_j\left(\frac{e^{\langle a,x\rangle}\cdot(1+e^{\langle a,x\rangle}) - e^{\langle a,x\rangle}\cdot e^{\langle a,x\rangle}}{(1+e^{\langle a,x\rangle})^2}\right) =$$

$$= a_i a_j \frac{e^{\langle a,x\rangle}}{(1+e^{\langle a,x\rangle})^2}$$

$$\nabla^2 f(x) = \frac{e^{\langle a,x\rangle}}{(1+e^{\langle a,x\rangle})^2} aa^T$$

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Задача № 4

Найти x : $\nabla f(x) = 0$

$$f(x) = \langle c, x \rangle + \frac{\sigma}{3} \|x\|^3$$

$$\nabla \|x\|^3 = 3\|x\|^2 \cdot \nabla \|x\| = 3\|x\|^2 \cdot \frac{x}{\|x\|}$$

$$\nabla f(x) = c + \frac{\sigma}{3} \cdot 3 \cdot \|x\| \cdot x = c + \sigma \|x\| x$$

$$\nabla f(x) = 0 \iff c = -\sigma \|x\| x$$

сумма двух векторов $= 0 \Rightarrow c \uparrow\downarrow -\sigma\|x\|x$ ($c \neq 0$)

$$\Rightarrow x \uparrow\uparrow c \Rightarrow x = -tc \; : \; t > 0$$

Осталось найти длины:

$$c = -\sigma \|(-tc)\| (-tc) = \sigma t^2 \|c\| \cdot c$$

$$\Rightarrow t^2 = \frac{1}{\sigma \|c\|} \Rightarrow t = \frac{1}{\sqrt{\sigma \|c\|}} \quad \text{т.к } t > 0$$

$$\Rightarrow \underline{x_0 = -\frac{c}{\sqrt{\sigma \|c\|}}}$$

Задача № 5

$$f(x) = \langle x, x \rangle^{\langle x, x \rangle} \qquad \exists f(x) = y$$

$$y = \langle x, x \rangle^{\langle x, x \rangle}$$

$$\ln y = \langle x, x \rangle \ln \langle x, x \rangle$$

$$\nabla \frac{y}{y} = 2x \cdot \ln \langle x, x \rangle + \langle x, x \rangle \cdot \frac{2x}{\langle x, x \rangle} = 2x (\ln \langle x, x \rangle + 1)$$

$$\underline{\nabla f(x) = 2x \cdot \langle x, x \rangle^{\langle x, x \rangle} \cdot (\ln \langle x, x \rangle + 1)}$$

$f(x) = x^x = y$

$\ln y = x \ln x$

$\frac{y'}{y} = \ln x + 1$

$y' = (\ln x + 1) x^x$

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$$\frac{\partial^2 f}{\partial x_i \partial x_j} = 2 \cdot \langle x,x \rangle^{\langle x,x \rangle} \cdot \left(\ln \langle x,x \rangle + 1\right) \gamma_{ij} +$$

$$+ 4 x_i x_j \cdot \langle x,x \rangle^{\langle x,x \rangle} \cdot \left(\ln \langle x,x \rangle + 1\right)^2 +$$

$$+ 2 x_i \cdot \langle x,x \rangle^{\langle x,x \rangle} \cdot \frac{2 x_j}{\langle x,x \rangle} =$$

$$\begin{cases} \gamma_{ij} = 1 & i = j \\ \gamma_{ij} = 0 & i \neq j \end{cases}$$

$$= 2 \langle x,x \rangle^{\langle x,x \rangle} \left(\ln \langle x,x \rangle + 1\right) \gamma_{ij} + 4 x_i x_j \langle x,x \rangle^{\langle x,x \rangle} \left(\left(\ln \langle x,x \rangle + 1\right)^2 + \frac{1}{\langle x,x \rangle}\right)$$

$$\nabla^2 f(x) = 2 \langle x,x \rangle^{\langle x,x \rangle} \left(\ln \langle x,x \rangle + 1\right) I + 4 \langle x,x \rangle^{\langle x,x \rangle} \left(\left(\ln \langle x,x \rangle + 1\right)^2 + \frac{1}{\langle x,x \rangle}\right) x x^T$$

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